India's first major Outsourced Semiconductor Assembly and Test (OSAT) facility in Jagiroad in 2024. That decision was as much a leap of faith as it was a meticulously planned manoeuvre. Tata's semiconductor packaging unit in Assam was the first of its kind in the Northeast, signalling that the industrial map of the region was no longer confined to tea, petroleum, and tourism. The OSAT plant was a behemoth—spread over two million square feet, powered entirely by green energy, and designed to produce over 15 billion chips annually. In a state where job creation had historically struggled to keep pace with its workforce, the Tata facility alone was projected to employ over 30,000 people directly and thousands more indirectly.

The decision to establish a semiconductor industry in Assam was not without its challenges. Historically, India's semiconductor investments have been clustered in southern and western states, where established ecosystems, supply chains, and infrastructure made manufacturing viable. Assam, with its relative industrial nascency, had to carve a path through uncharted territory. The state government, under Sarma's stewardship, rolled out a slew of incentives to make the venture attractive—tax breaks, infrastructure upgrades, and a robust skilling program in collaboration with Indian Institute of Technology, (IIT) Guwahati and National Institute of Electronics & Information Technology (NIELIT).

With the Tata plant operational and Advantage Assam 2.0 adding an influx of international investment, Assam's semiconductor roadmap is now charting an ambitious future. Five semiconductor projects in India with a cumulative investment of Rs 1.5 lakh crore are under construction which will further deepen Assam's integration into the global semiconductor supply chain. Near to the OSAT plants, the state is setting up a Greenfield Electronics Manufacturing Cluster with a focus on mobile phone and telecom component manufacturing. These projects are not just about infrastructure—they are about positioning Assam as a serious contender in India's semiconductor policy, aligning seamlessly with the national India Semiconductor Mission that aims to make the country self-reliant in chip manufacturing.

Beyond its obvious economic implications, Assam's semiconductor ascent is poised to revolutionize the state's job market and academic institutions. The direct employment potential runs into tens of thousands, but the indirect impact is even more transformative. New suppliers and ancillary industries—precision tooling, cleanroom equipment, packaging materials—will see a boom, while the demand for electrical engineers, material scientists, and chip designers is set to create a new knowledge economy in the region. IIT Guwahati, NIT Silchar, and Gauhati University have already started tailoring their curriculum to align with semiconductor research, with specialized programmes in chip design and fabrication.



THE SEMICONDUCTOR JOURNEY

- Advantage Assam 2.0 secured MoUs worth Rs 5 lakh crore, with 10 signed with leading semiconductor firms from Singapore, Malaysia, and Japan
- Five semiconductor projects worth Rs 1.5 lakh crore under development.
- Rs 27,000 crore investment in Tata Electronics' OSAT facility at Jagiroad, producing 48 million chips daily.
- Direct and indirect employment in semiconductors in tens of thousands, transforming Assam's economy. The Tata plant will employ 30,000+ people.
- Assam's youth trained in Taiwan for advanced chip-packaging skills.
- IIT Guwahati, NIT Silchar, and Gauhati University launching semiconductor-focused programmes to skill the youth
- Assam offering tax breaks, infrastructure upgrades, and policy support for semiconductor investments.

IIT Guwahati and the Assam Government are working on a long-term partnership to skill youth from Assam and its neighboring states in semiconductors, aiming to create adequate manpower to meet industry demands. IIT Guwahati is actively shaping India's semiconductor ecosystem through research, training, and industry collaborations. As an INUP (Indian Nano Users Program) partner, it offers NSQF-certified skilling courses in nanotechnology, semiconductor packaging, and industrial automation, with hands-on training and clean room access. It has 58+ global research collaborations, including 18+ in Japan, and runs joint Ph.D. and Master's programmes in Materials Science, Quantum Mechanics, and Device Fabrication. A tripartite partnership with TATA OSAT and NIELIT is driving workforce development, particularly for women and youth. Additionally, IIT Guwahati is fostering a startup ecosystem in semiconductors while managing the INUP-i2i central web portal for nationwide research access. This fusion of academia and industry will not only ensure that Assam retains its best talent but will also make it a magnet for researchers and professionals from across India.

For Assam, the semiconductor industry is not just an economic booster; it is a statement of intent. A state long seen as peripheral to India's industrial ambitions is now defining a new frontier. In a world increasingly dependent on semiconductor technology—where every device, every car, every aspect of modern life hinges on the availability of microchips—Assam's positioning could not be more strategic. And for Dr Sarma, the sleepless nights, the manic schedules, and the relentless lobbying have not just paid off—they have altered the very contours of Assam's future. ▪